

Fostering technical proficiency and professional skills: A multifaceted PO-PBL strategy for unit operations education

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ABSTRACT

This contribution presents a comprehensive analysis of the implementation and outcomes of a Project-Oriented Problem-Based Learning (PO-PBL) approach in chemical engineering education, focusing on the Project of Unit Operations (POU) course. The study examines the efficacy of PO-PBL in developing both technical proficiency and essential professional skills among students. Using a mixed methods approach, the research analyzes data from four consecutive semesters, encompassing student surveys, performance metrics, and qualitative feedback. The integration of the Integrated Product and Process Design (DIPP) methodology, gamification elements, and Generative Artificial Intelligence (GenAI) tools are explored as complementary strategies to improve learning outcomes. The results indicate sustained or improved technical skills throughout the semesters, coupled with significant improvements in student engagement, motivation, and professional competencies, particularly in communication and teamwork. The implementation of GenAI tools, guided by the Artificial Intelligence Assessment Scale (AIAS), shows promising improvements in students' research capabilities and writing skills. Although challenges in implementation are acknowledged, the POU course emerges as a promising model for chemical engineering education, effectively bridging the gap between theoretical knowledge and practical application. This study provides valuable information on innovative pedagogical approaches that prepare students for the evolving demands of the chemical engineering industry.

1. Introduction

Chemical engineering education is currently experiencing a pivotal transformation, necessitated by the imperative to adapt to the unique learning styles characteristic of Generation Z. This cohort resides in an environment where digital and physical domains converge seamlessly. These individuals, molded by incessant connectivity and rapid technological advancements, present challenges to conventional educational methodologies, thereby demanding strategies that align with their distinctive cognitive processes (Hendrastomo and Januarti, 2023; See-miller and Grace, 2016).

Traditional methods in chemical engineering instruction—lectures, problem-solving exercises, and structured lab work—though foundational, increasingly fall short in engaging students and developing the

multifaceted skills needed for navigating complex industrial landscapes.

Recent advances in chemical engineering education have introduced novel methodologies that complement traditional approaches. Virtual and augmented reality have emerged as powerful tools for visualizing complex processes (Kumar et al., 2021; Gunawan et al., 2021), while data-driven approaches are revolutionizing how students interact with process simulation (Naidoo et al., 2023). Integrating artificial intelligence into engineering education has opened new possibilities for personalized learning and automated feedback systems (Iyamuremye et al., 2024). These innovations, combined with established methodologies such as PBL, reshape how future chemical engineers are trained.

Faced with these educational gaps, innovative teaching strategies, such as problem-based learning (PBL), are gaining traction as a way to bridge the gap between theory and practice. PBL immerses students in

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complex real-world problems from day one, transforming them from passive recipients of information into active participants in their learning journey (Robinson and Beneroso, 2022; Ricaurte and Viloria, 2020). Based on constructivist learning theory, PBL suggests that students construct knowledge through experience and reflection, not merely by absorbing data from lectures (Schmidt et al., 2009).

At the Universidad de los Andes, we have embraced this pedagogical shift, weaving PBL into our chemical engineering courses. Our initial implementation in the Unit Operations course challenged the students with designing, 3D printing, and analyzing a centrifugal pump (Ballesteros et al., 2019). Although practically relevant, students struggled to connect this task with broader course content, and time constraints limited the deeper exploration of design considerations.

Recognizing these challenges, we refined our approach through flipped classrooms, diverse PBL tasks, and interactive e-learning tools to enhance engagement and learning outcomes (Ballesteros et al., 2021). Despite these improvements, there were opportunities to further develop advanced problem-solving and innovative thinking skills critical to engineering practice.

In response to the challenges posed by traditional teaching methods, we decided to adopt the Project-Oriented Problem-Based Learning (PO-PBL) approach, a well-established pedagogical strategy that integrates project-driven learning experiences into the curriculum. Drawing on situated learning theory and the concept of cognitive apprenticeship, PO-PBL enables educators to serve as mentors, guiding students through progressively complex tasks (Brown et al., 1989; Collins et al., 1991). Our course, titled Project of Unit Operations (POU), leverages the PO-PBL framework to offer students long-term projects that span the semester, allowing them to apply their knowledge of unit operations within an industrially simulated environment.

This contribution examines the implementation and results of the PO-PBL model within the POU course. Through a mixed-method analysis of four semesters' worth of data, including student surveys, performance metrics, and qualitative feedback, we explore the impact of the PO-PBL framework and its integration with the DIPP methodology, gamification, and GenAI tools. We hypothesize that this innovative course design will deepen students' understanding of unit operations while developing crucial problem-solving and communication skills essential for success in the chemical engineering field.

We hypothesize that this innovative course design will not only deepen the students' comprehension of unit operations but also sharpen their problem-solving skills and communication skills, key elements for success in the chemical engineering field. This research offers meaningful insights into cutting-edge pedagogical strategies that align chemical engineering education with the demands of a rapidly evolving industry.

2. The integrated product and process design (DIPP) methodology

The Integrated Product and Process Design (DIPP, from its Spanish acronym) methodology brings a fresh lens to chemical engineering education by weaving together product design and process engineering into a cohesive framework. As described by Alvarez (2017), DIPP bridges two traditional but often separate viewpoints: one that sees process design as the central focus with the product being just another component and the other that prioritizes the characteristics of the product, viewing the process simply as a means to achieve the desired result.

What sets DIPP apart is its unified approach, which marries the manufacturing process with the product's formulation and properties. At the heart of this methodology lies the multiscale “3 P” tool (Product, Process, Properties), which establishes critical links between these three domains Alvarez (2017). This holistic perspective not only enriches the educational experience but also enhances practical applications, providing students with a comprehensive understanding of both product

development and process design.

2.1. Steps in the DIPP methodology

The DIPP methodology unfolds through a series of adaptable steps, each aimed at systematically identifying and addressing opportunities and needs (see Fig. 1).

The strategy begins by identifying potential opportunities or unmet needs and then moves into a vigorous phase of idea generation and selection. Here, the multiscale “3 P” tool steps in, facilitating the conversion of these ideas into tangible prototypes. By linking the properties of the product on different scales with the manufacturing conditions and the formulation of the product, the tool ensures a comprehensive analysis of each design (Pradilla et al., 2015; Calvo et al., 2020; Díaz-Bustamante et al., 2022). The methodology then shifts focus toward integrating sustainability principles, ensuring that the final product not only meets current needs but also remains viable and relevant for the future. The implementation phase culminates in the implementation of the design and the guarantee of its feasibility and effectiveness in real-world applications.

2.2. DIPP and creativity

Creativity is inherently woven into the DIPP methodology. This synergy can be examined through the “4Ps” framework of creativity: Person, Product, Process, and Press (Rhodes, 1961; Cropley, 2015). Building on this, Cropley (2015) introduced a seven-stage model of creativity, emphasizing the interplay of convergent (focused) and divergent (exploratory) thinking throughout the process. Table 1 offers a comparative analysis of these creative stages with the steps of the DIPP methodology and the popular *Design Thinking* approach (Brockman, 2009; Horenstein, 2002).

By integrating creativity, product design, and process development, DIPP offers a robust framework that equips chemical engineers with the tools to develop innovative solutions. This methodology ensures that engineers not only meet current industrial demands but also anticipate future challenges by considering the intricate relationships between product properties, manufacturing processes, and sustainability.

3. Course design and implementation

3.1. Overview of the POU course structure

The Project of Unit Operations (POU) is a 16-week course that integrates four key competencies: *Concepts to Practice*, *Problem Solving*, *Communication Skills*, and *Teamwork* (Fig. 2A). The course employs a Project-Oriented Problem-Based Learning (PO-PBL) approach, enhanced by Generative Artificial Intelligence (GenAI) tools and gamification elements to develop both technical expertise and professional skills.

The course structure comprises three main components with a total of 6 weekly contact hours. The project sessions, conducted twice weekly for a total of 3 h, focus on project development, team meetings, and milestone presentations. These sessions facilitate expert consultations and peer reviews while monitoring the progress of the project. Weekly 1.5-h laboratory sessions provide hands-on experimentation with unit operations equipment, emphasizing data collection and analysis through six structured practices that focus on fundamental operations such as filtration, distillation, and heat exchange. Complementary sessions, also 1.5 h weekly, deliver theoretical workshops covering specific unit operations topics, process flow diagram development, and specialized training in technical communication. Beyond scheduled contact hours, students dedicate approximately three hours weekly to independent project work, research, and deliverable preparation.

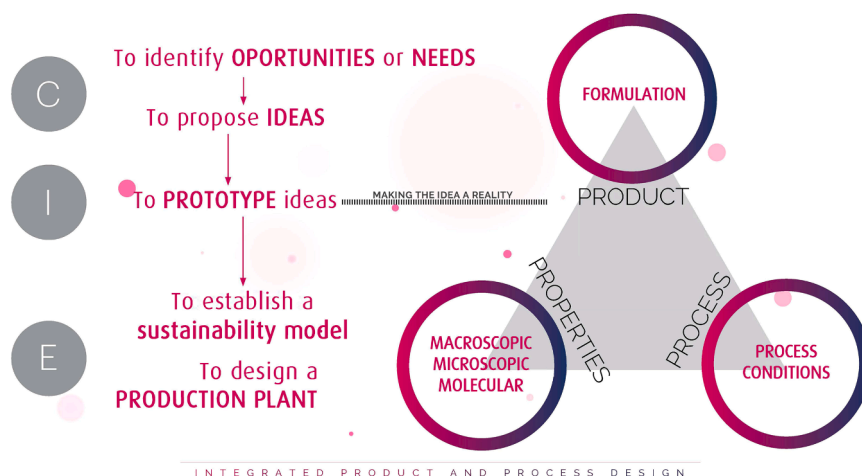


Fig. 1. The Integrated Product and Process Design (DIPP) Methodology.

Table 1

Comparison of the creative process with the DIPP and the Design Thinking methodology.

Creative Process	DIPP	“Design Thinking”
Preparation	Identify opportunities or needs	Understand
Activation	Generate and select the best ideas	Observe
Generation		Visualize
Illumination	Prototype and ensure sustainability	Prototype
Verification		
Communication	Test and refine the prototype	Test the design
Validation	Implement the final design	Implement

3.2. Project development and assessment

The course focuses on a semester-long project that progresses through key milestones (Fig. 2B). The journey begins with a Design Assignment, where students receive a unique industry-based challenge (Supplementary information 2). Teams then move through an Ideation Phase, presenting initial solutions through professional flyers (Supplementary information 3), followed by Disciplinary Challenges that test the application of theoretical concepts (Supplementary information 4). As the semester progresses, Adaptability Challenges evaluate teams' ability to modify solutions under changing conditions (Supplementary information 5). The project culminates in a prototyping phase and final presentation, where the teams showcase their work through comprehensive technical documentation and poster sessions (Supplementary information 6, Supplementary information 7).

The assessment follows a continuous model throughout the semester. The project milestones account for 40% of the final grade, encompassing the initial design (5%), flyer development (5%), the disciplinary challenges (15%), and the adaptability challenges (15%). Laboratory performance and reports contribute 20%, while final project deliverables make up the remaining 40%, distributed throughout the technical document (20%), poster presentation (10%), and final defense (10%).

3.3. Communication skills development

The development of professional communication is a fundamental component of the course. The *Centro de Español* (Universidad de los Andes, 2024) provides specialized support for technical writing through workshops and online modules with associated quizzes (Supplementary information 10). Complementing this writing focus, the teaching team of the course delivers comprehensive visual communication training, equipping students with skills in data visualization, analysis, and the creation of industry-standard diagrams (Supplementary information

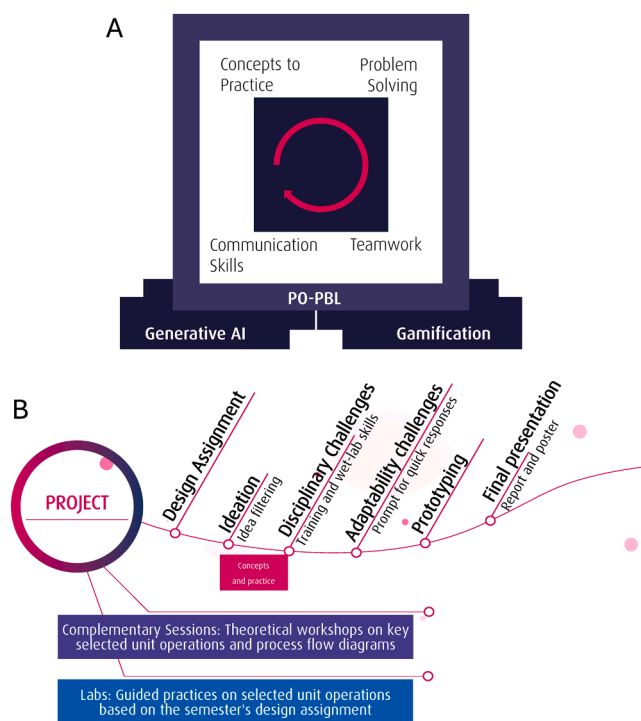


Fig. 2. POU course structure. (A) Competencies developed through the PO-PBL approach. At its core, the model emphasizes four interconnected competencies: Concepts to Practice, Problem Solving, Communication Skills, and Teamwork. (B) Project milestones and supporting activities throughout the semester. The process begins with a Design Assignment and progresses through stages of Ideation, Disciplinary Challenges, Adaptability Challenges, and Prototyping, culminating in a Final Presentation. Supporting this core process are two complementary components: theoretical workshops focusing on key unit operations and process flow diagrams and laboratory sessions offering guided practices on selected unit operations.

11). This dual approach, which combines expert writing instruction with professional visual communication training, ensures students develop the comprehensive communication skills essential for success in the chemical engineering field.

3.4. Innovative learning elements

The POU course distinguishes itself through three innovative

elements that enhance the learning experience. First, Adaptability Challenges present real-time problem-solving scenarios that test students' ability to modify their solutions under changing conditions. Second, structured integration of GenAI tools improves research capabilities and technical writing skills (Supplementary information 12). Third, a comprehensive gamification system featuring point tracking, leaderboards, and achievement recognitions increases engagement and motivation throughout the semester. This integrated approach ensures that students develop both technical mastery and essential professional skills, effectively preparing them for chemical engineering careers.

3.5. Integration of PO-PBL principles

The POU course deeply integrates PO-PBL principles, creating a learning environment mirroring real-world engineering challenges. Central is a semester-long project that evolves through distinct phases to cultivate essential skills. The course starts with a *Design Assignment* introducing a unique fictitious industry scenario each term, crafted by the professor. This engages students and challenges them to adapt, mirroring real-world adaptability. Students face industry-relevant challenges in diverse applications, from natural flavorings to antifungal patches (see Table 2). This exposure familiarizes them with various sectors of the chemical industry and highlights their versatility.

A two-digit code following the year designates the semesters in our institution's academic calendar:

- 10 indicates the first semester (January to June).
- 20 indicates the second semester (July to December).

For example, 2023–10 refers to the first semester of 2023. Recent iterations of the POU course demonstrate a deliberate progression in challenge complexity and scope. In 2022–20, with EcoFood, students focused on distillation processes for commercial products, requiring mastery of separation techniques and process optimization. The Bio-lógicos challenges in 2023–10 expanded this foundation by introducing multiple product lines across the cosmetic, food, and pharmaceutical sectors, requiring students to consider diverse regulatory requirements and cross-industry applications. The 2023–20 BetaSolutions projects centered on betanin applications, which required students to adapt a single bioactive compound for different industrial contexts while considering stability challenges and formulation requirements. The most recent iteration (2024–10) with BioInnova represents the highest complexity level, challenging students to develop an antifungal patch incorporating Colombian biodiversity. This latest challenge integrates multiple unit operations, from extraction to polymer processing, while

Table 2
Challenges presented in recent POU course iterations.

Semester	Fictitious Company	Challenge	Technical Focus
2022–20	EcoFood	Design a production process for new distillates to expand the company's commercial portfolio.	Separation processes, thermal unit operations, process optimization
2023–10	Bio-logicos	Design production processes for natural flavorings, food additives, and antioxidant extracts.	Extraction techniques, purification methods, stability studies
2023–20	BetaSolutions	Design production processes for betanin-based products across food, cosmetic, and textile applications.	Solid-liquid extraction, powder processing, pigment stability
2024–10	BioInnova	Develop an antifungal patch using bioactive components from Colombian endemic biodiversity.	Bioactive extraction, polymer processing, controlled release systems

addressing biocompatibility and drug delivery considerations. Each iteration builds on previous experiences, progressively increasing the integration of technical knowledge with practical industry constraints.

Students proceed through activities: team formation, initial ideation, participation in expert communities, and face *adaptability challenges* that push them to refine ideas in response to changes. They develop communication materials, such as flyers, that bridge technical and professional communication skills.

A key strength is the emphasis on prototyping, where students turn ideas into tangible realities, applying theory in hands-on scenarios, fostering creativity and problem-solving. The project culminates in a *Final Defense*, with students presenting their solutions in a document and poster, reflecting the expectations of the professional industry.

Students drive their learning journey with guidance from professors and TAs, but final decisions are made by teams, cultivating ownership and responsibility, which are key PO-PBL tenets.

Reflection and self-assessment are integral. Weekly team meetings for project logs encourage reflection. Self- and peer-evaluations, based on team-established criteria (Supplementary information 13), promote accountability and self-assessment, aligning with psychological safety and team norms.

Though primarily for chemical engineering students, the course encourages interdisciplinary thinking by seeking insights from related fields like food engineering, product design, and biomedical engineering. This broadens perspectives and prepares students for interdisciplinary challenges. Integrating practical problem-solving, prototyping, self-directed learning, and reflection, the POU course creates a multifaceted learning experience, developing technical knowledge and essential professional skills, embodying PO-PBL's essence.

3.5.1. Course learning outcomes

The POU course is designed to achieve the following learning outcomes.

1. LO1: Identify and analyze industry challenges within the chemical engineering field.
2. LO2: Apply creative problem-solving techniques to generate innovative solutions.
3. LO3: Integrate the principles of unit operations into the process design.
4. LO4: Demonstrate adaptability in dynamic scenarios and refine processes accordingly.
5. LO5: Apply multiscale analysis to product development, considering molecular to macroscopic properties.
6. LO6: Evaluate the environmental and social impacts to ensure the sustainability of processes.
7. LO7: Communicate technical information effectively through written, visual, and oral presentations.

3.6. Incorporation of gamification elements

The POU course incorporates gamification to boost engagement and motivation, using a comprehensive point system for assignments, milestones, and activities, turning them into opportunities to accumulate points and track progress.

The point system reflects the importance and complexity of course components: major deliverables like the final poster and document carry 200 points each; quizzes award 5 points; tasks like disciplinary challenges can earn up to 100 points. This encourages participation in all aspects of the course.

A key gamification feature is the publicly visible leaderboard, updated weekly via email and the LMS, fostering competition and allowing students to gauge their progress, serving as a motivational tool.

Gamification includes bonus points: at the semester's end, 100 extra points are awarded across categories like *best presentation* and *best project*. In a recent iteration, 100 extra points were awarded to the team

that achieved the most significant inhibition halo against *Candida tropicalis*, encouraging excellence and bridging theory with practice.

During the final session, all visitors to the stands evaluate significant deliverables, such as the final poster and presentation, to ensure fairness and thorough evaluation. This includes several graduate assistants, professors, and sometimes industry professionals, resulting in up to 20 evaluations per team. Each evaluator uses a standardized rubric focusing on criteria such as technical content, clarity, innovation, and visual appeal.

The final grade for these deliverables is determined by averaging the scores from all evaluators. This collective assessment approach provides diverse feedback and ensures that the assessment is balanced and representative of different perspectives.

In addition, awards are awarded for the best poster and presentation based on these evaluations. The high number of evaluators not only enhances the reliability of the assessment but also exposes students to a broader audience, simulating real-world professional presentations.

The effectiveness of the gamified approach is evident in student engagement. With more than 1200 points available and a maximum grade of 1000 points, students are motivated to exceed the requirements. This surplus system allows personal goal setting and buffers setbacks, reducing stress. Research supports point-based reward systems to improve motivation and engagement [Derado et al. \(2015\)](#), encourage consistent study habits, and offer multiple ways to demonstrate understanding [Ching \(2012\)](#).

Importantly, gamification complements rather than overshadows learning objectives, enhancing motivation and engagement with the material.

No observed drawbacks suggest successful implementation and positive student reception, balancing healthy competition and collaboration essential for team-based projects.

By integrating gamification, the POU course creates a dynamic environment, motivating students to excel in understanding and applying chemical engineering principles, and enhancing traditional education with modern engagement strategies.

3.7. Use of GenAI tools in the course

Since the second semester of 2023, the POU course has integrated GenAI tools, recognizing their growing importance, enhancing learning while maintaining academic integrity, and fostering critical thinking. Of the four semesters discussed, the GenAI tools were integrated starting from the second semester of 2023 (2023–20). The semesters 2022–20 and 2023–10 were conducted without GenAI assistance.

To guide students in the ethical use of GenAI tools, we implemented the Artificial Intelligence Assessment Scale (AIAS) [\(Perkins et al., 2024\)](#). This framework provided clear guidelines on the acceptable usage of GenAI, helping students understand what was expected regarding the use of these tools.

We trusted our students to adhere to these guidelines, emphasizing the importance of academic integrity and responsible use of AI. To assess their understanding and ensure that they engaged deeply with the material, we relied on follow-up through experimental work and oral presentations. During laboratory sessions, students performed experiments and collected data first-hand, requiring active participation and comprehension of the procedures. In oral presentations, the students explained their projects, defended their methodologies, and responded to questions from the instructors and peers. This interactive setting made it clear whether the students had genuinely engaged with the references and understood the content, as they needed to articulate their thoughts and demonstrate critical thinking. By observing students in these settings, we could effectively assess their mastery of the subject matter, ensuring that even if GenAI tools were used for assistance, the students had internalized the knowledge and could apply it appropriately.

Effective integration relies on the instructor's proficiency with GenAI tools. The professor has extensive training, allowing nuanced guidance

on capabilities, limitations, and ethical considerations.

The course cultivates responsible use of GenAI through structure, guidelines, and discussions, emphasizing technical aspects and critical thinking. Students receive best-practice handouts and must acknowledge the use of GenAI in their work.

The comparative analysis of literature reviews across semesters examined the impact of specialized GenAI research tools on review quality. Unlike traditional large-language models, the tools used, Perplexity [\(Perplexity AI, 2023\)](#), SciSpace [\(Khan et al., 2018\)](#), and Consensus [\(Consensus, 2024\)](#), are specifically designed for academic research with direct access to scientific databases and citation networks. Perplexity connects to academic repositories in real time, providing current research findings and citations. SciSpace specializes in analyzing scientific papers, offering detailed insights into methodologies and findings. Consensus synthesizes research papers from peer-reviewed sources, helping to identify key trends and findings across multiple studies.

Students used these specialized tools for literature discovery and synthesis, while reserving ChatGPT® [\(OpenAI, 2024\)](#) for broader ideation and writing refinement tasks. This deliberate separation ensured that the literature reviews remained grounded in verifiable academic sources rather than AI-generated content. The analysis compared the performance of students before and after the introduction of these research-specific AI tools, examining metrics such as the number of high-impact references, the accuracy of citations, and the temporal distribution of sources [\(Table 3\)](#).

By thoughtfully integrating GenAI, the POU course prepares students for a future where AI is integral, equipping them to harness these tools while maintaining rigor and creativity. The instructor's proficiency models an effective and ethical integration into professional practice.

3.8. Application of DIPP methodology in the course

The POU course combines PO-PBL with the DIPP methodology, providing a structured framework for tackling challenges. While PO-PBL addresses the *what* of learning objectives, DIPP offers a systematic *how* for problem-solving and design, emphasizing the integration of product and process development.

Each course activity is strategically designed to align with the DIPP methodology and the course learning outcomes defined earlier. [Table 4](#) presents this alignment.

Each activity not only progresses students through the DIPP methodology but also systematically achieves the intended learning outcomes of the course.

For instance, during the *Design Assignment* phase, the professor introduces the students to a distinctive fictitious industry scenario. This

Table 3
Alignment of course guides with AIAS levels.

Guide	AIAS Level	Justification
Pre-writing: Generating research questions and ideas	Level 2: AI-Assisted Idea Generation and Structuring	Students use AI to brainstorm and structure ideas, but the final content is their own.
Literature review using Perplexity, SciSpace, and Consensus	Level 3: AI-Assisted Editing	AI tools help summarize and analyze existing literature, but students must synthesize and write their own reviews.
Methodology development assistance	Level 2: AI-Assisted Idea Generation and Structuring	AI suggests improvements to methodology, but students make final decisions and write the content.
Creating outlines for research articles	Level 2: AI-Assisted Idea Generation and Structuring	AI helps structure the article, but content creation remains the student's responsibility.
Post-writing: Revision and editing	Level 3: AI-Assisted Editing	AI assists in refining language and structure, but original content is preserved.

Table 4
Mapping of course tasks to DIPP steps and course learning outcomes.

Course Task/Activity	DIPP Step	Course Learning Outcome(s)
Design Assignment	Identify opportunities/needs	LO1: Identify and analyze industry challenges.
Ideation Phase (Flyers)	Generate and select ideas	LO2: Apply creative problem-solving techniques.
Disciplinary Challenges	Develop prototypes using 3 P tool	LO3: Integrate unit operations into process design.
Adaptability Challenges	Refine prototypes and processes	LO4: Demonstrate adaptability in dynamic scenarios.
Prototyping Phase	Multiscale characterization	LO5: Apply multiscale analysis to product development.
Sustainability Analysis	Consider sustainability aspects	LO6: Evaluate environmental and social impacts.
Final Presentation (Poster and Document)	Implement and communicate solutions	LO7: Communicate technical information effectively.

aligns with the first step of DIPP *identify opportunities/needs* and fulfills LO1: *Identify and analyze industry challenges within the field of chemical engineering*. Students face industry-relevant challenges in diverse applications, from natural flavorings to antifungal patches (see Table 2). This exposure familiarizes them with various sectors of the chemical industry and highlights the versatility of their skills.

In the *Ideation Phase*, students employ creative thinking to generate innovative solutions, directly mapping to DIPP's *Generate and select ideas* step and fulfilling LO2: *Apply creative problem-solving techniques to generate innovative solutions*. This phase encourages divergent thinking as students brainstorm various possibilities and convergent thinking as they evaluate and select the most promising ideas using tools like SWOT analysis and Pugh matrices.

The *Disciplinary Challenges* require students to apply principles of unit operations within the context of process design, integrating theoretical knowledge with practical application. This aligns with DIPP's *Develop prototypes using the 3 P tool* step and addresses LO3: *Integrate principles of unit operations into process design*. By tackling real-world engineering problems, students learn how individual unit operations contribute to the overall process and how to optimize them for desired outcomes.

During the *Adaptability Challenges*, students refine their prototypes and processes in response to new constraints or information, reflecting DIPP's emphasis on iterative development. This fulfills LO4: *Demonstrate adaptability in dynamic scenarios and refine processes accordingly*. These challenges simulate unexpected changes common in engineering projects, such as resource limitations or changing client requirements, teaching students to adjust their designs efficiently.

The *prototyping phase* involves multiscale characterization of their products, from molecular to macroscopic properties, corresponding to DIPP's focus on multiscale analysis. This addresses LO5: *Apply multiscale analysis to product development, considering molecular to macroscopic properties*. For example, in developing an antifungal patch, students used techniques from FTIR and IR for molecular analysis to texturometry and rheology for macroscopic characterization, even conducting antibiograms on pathogenic yeast, understanding how properties at different scales influence the final product's performance.

In the *Sustainability Analysis*, students evaluate the environmental and social impacts of their processes, aligning with the DIPP's consideration of sustainability aspects. This fulfills LO6: *Evaluate environmental and social impacts to ensure the sustainability of processes*. They write argumentative texts analyzing how their designs affect sustainability goals, prompting them to consider waste reduction, energy efficiency, and social responsibility.

Finally, the *Final Presentation* (poster and document) requires students to implement and communicate their solutions effectively,

mirroring the DIPP implementation step. This addresses LO7: *Communicate technical information effectively through written, visual, and oral presentations*. Students synthesize their findings and present them to an audience of peers, instructors, and industry professionals, improving their ability to convey complex technical concepts clearly and professionally.

By integrating the DIPP methodology into the course structure, students gain hands-on experience applying chemical engineering principles to real-world challenges. This approach not only imparts technical knowledge but also cultivates creative, analytical, and communication skills, preparing students to tackle complex engineering challenges holistically.

Reflection and self-assessment are integral components of the course. Weekly team meetings for project logs encourage reflection, while self and peer evaluations, based on team-established criteria (Supplementary information 13), promote accountability and self-assessment. This aligns with psychological safety and team norms, further reinforcing the emphasis of the DIPP methodology on continuous improvement and learning.

Though primarily for chemical engineering students, the course encourages interdisciplinary thinking by seeking insights from related fields like food engineering, product design, and biomedical engineering. This broadens perspectives and prepares students for interdisciplinary challenges. By integrating practical problem-solving, prototyping, self-directed learning, and reflection, the POU course creates a multifaceted learning experience that embodies the essence of both PO-PBL and DIPP methodologies.

4. Research questions and methodology

4.1. Key research questions

This study aimed to evaluate the effectiveness of the PO-PBL approach, integrated with the DIPP methodology, in developing a comprehensive set of competencies essential for success in chemical engineering. The research focused on the following key questions.

1. How effectively does the PO-PBL methodology enhance students' understanding of chemical engineering principles, particularly unit operations, through hands-on problem-solving?
2. How does the combination of PO-PBL, GenAI tools, and gamification elements influence students' engagement and motivation in learning chemical engineering concepts?
3. What impact does this innovative course structure have on developing students' communication skills, both written and visual?
4. How does the PO-PBL approach contribute to the development of professional skills such as teamwork, creativity, innovation, and entrepreneurship?
5. What effect does the integration of GenAI tools have on students' research capabilities and writing skills, particularly in conducting literature reviews and crafting methodologies?
6. To what extent does the POU course, with its multifaceted approach combining PO-PBL and DIPP, equip students for the diverse challenges awaiting them in the chemical engineering industry?

4.2. Description of the cohorts for analysis

The study examined four diverse student cohorts that span consecutive semesters. The 2023–10 cohort had 32 students (19 male, 13 female), coming in second to the 2022–20 cohort with 38 students (22 male, 16 female). The 2023–20 cohort consisted of 18 students (3 male, 15 female), and the most recent 2024–10 cohort included 24 students (17 male, 7 female). This broad sample allowed us to track trends and make longitudinal comparisons between various aspects of the course.

4.3. Description of the student survey

A detailed survey was administered at the end of the POU course (Supplementary information 14), focusing on the 2024–10 cohort. With 19 of 24 students participating, achieving a response rate of 79.2%, the survey provided a representative snapshot of student perspectives from the most recent iteration of the course. The survey explored key areas such as the perceived effectiveness of the PO-PBL methodology, the development of creativity and innovation skills, the advancements in visual and written communication, the experiences with GenAI tools for the review and ideation of the literature, the impact of hands-on activities and challenges, and the quality of feedback received throughout the course. To ensure a nuanced understanding of student experiences, the survey employed a mix of Likert scale questions for quantitative analysis and open-ended questions for qualitative insight.

4.4. Process of literature review comparison (with and without GenAI assistance)

A comprehensive comparative analysis was performed over two distinct semesters to assess the impact of GenAI tools on the quality of the literature review. This analysis compared the performance of students with and without access to GenAI tools, focusing on several key metrics. These included the average number of references used, the number of sources of high impact cited, the temporal distribution of the references, and the completeness and precision of the citations. This multifaceted approach aimed to determine whether GenAI tools led to more thorough, current, and accurately cited literature reviews, addressing a previously observed issue of superficiality in student research.

4.5. Data analysis methods

The methodology used a mixed methods approach that combined quantitative and qualitative data analysis. The quantitative analysis included descriptive statistics to summarize trends in student performance and perceptions, one-way ANOVA and Tukey's HSD tests to assess differences in communication skills across semesters, and t-tests to compare literature review metrics between semesters with and without GenAI tools. Qualitative analysis involved thematic analysis of open-ended survey responses and student feedback to identify recurring themes and insights on the course's impact on learning, skill development, and industry preparedness.

The study also incorporated longitudinal analysis, examining data over four semesters (2022–20–2024–10) to track changes and improvements over time, particularly in areas such as communication skills and technical proficiency. A comparative analysis between semesters was conducted to evaluate the impact of specific interventions, such as the introduction of GenAI tools and targeted communication skill development initiatives.

To address potential scoring biases, an independent assessor was employed to re-assess written communication skills using a standardized rubric, comparing documents from different semesters. This comprehensive methodological approach allowed for the triangulation of findings from multiple data sources, providing a robust evaluation of the effectiveness of the POU course in achieving its educational objectives.

5. Results

The evolution of teaching methodologies in our Unit Operations course offers valuable insights into how our current approach has shaped student outcomes. Previous implementations centered on flipped classroom models and smaller-scale PBL activities (Ballesteros et al., 2021). While these earlier strategies showed some promise in enhancing technical competencies, they were less effective in fostering meaningful peer interactions and sustaining student motivation. The predominant

focus on technological tools and flipped content delivery led to only moderate improvements in technical skill acquisition and yielded limited gains in professional skill development.

In contrast, our current PO-PBL implementation demonstrates marked improvements across multiple dimensions. In areas where the previous methodology struggled—such as promoting genuine peer-to-peer dialogue and cultivating emotional engagement (Ballesteros et al., 2021)—our updated approach has facilitated deeper involvement through structured teamwork and industry-relevant project challenges. The integration of adaptability tasks and a comprehensive gamification framework has particularly addressed the prior issue of waning engagement. Notably, student reports of consistent motivation throughout the semester increased from 67% under the previous approach to 94% with the current implementation.

Moreover, the incorporation of the DIPP methodology and GenAI tools has strengthened those domains that previously showed limitations. Although earlier approaches emphasized technical concept mastery, our current framework now supports balanced growth in both technical and professional competencies. This is especially evident in the improvement of communication skills. By integrating structured writing exercises and focused visual communication training, we have achieved statistically significant enhancements in students' communication performance. In sum, the current PO-PBL approach not only refines technical understanding but also enriches the collaborative, communicative, and adaptive skills essential for today's engineering practice.

5.1. Understanding of chemical engineering principles

The examination of technical skills over four semesters (2022–20–2024–10) reveals a steady level of understanding of chemical engineering principles among students. Fig. 3 shows the distribution of technical skills scores derived from evaluations of final project documents and weekly quizzes.

Analysis of the data reveals subtle fluctuations in performance across semesters. The median scores show a modest upward trend from 2022 to 2023–20, suggesting a gradual enhancement in general technical skills. Specifically, the median scores were 3.7, 3.8, 3.9, and 3.8 for the 2022–20, 2023–10, 2023–20 and 2024–10 semesters, respectively. The 2024–10 semester exhibits a wider interquartile range (3.5–4.2), indicating greater variability in student performance.

To probe deeper into the impact of practical activities on theoretical understanding, an analysis of students' responses regarding laboratory activities and specific challenges was conducted. Feedback strongly supports the efficacy of hands-on experiences in bridging the gap between theory and practice. 85% of the students reported that seeing theoretical phenomena in practical settings significantly boosted their grasp of complex concepts, particularly in understanding unit operations. One student aptly captured this value: “Visualizing in the laboratory

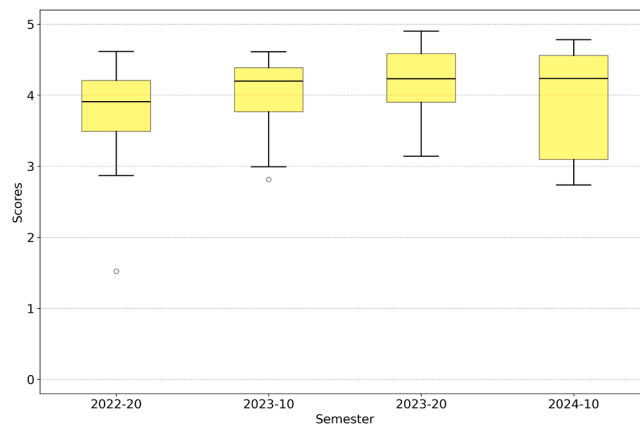


Fig. 3. Distribution of technical skills scores across semesters.

how things happen allows you to know how theory is applied in reality and the challenges this entails.” Another student noted, “The practical sessions helped me understand the theoretical concepts much better, especially when we encountered unexpected results and had to troubleshoot.”

The practical sessions not only reinforced technical knowledge but also nurtured critical professional skills. 78% of the students reported improved equipment handling proficiency, and 72% noted a more nuanced understanding of processes. One student remarked, “The lab work really challenged us to think on our feet and apply our knowledge in real time. It is one thing to understand a concept in theory but quite another to make it work in practice.”

However, an area for potential improvement was noted, and students indicated a disconnect between course components. One student pointed out that “...the part of the course focused on unit operations is still very disconnected from the project part and laboratory practices.” This feedback suggests an opportunity for further integration of theoretical and practical elements in future course iterations.

5.2. Student engagement and motivation

The implementation of the PO-PBL methodology has significantly boosted student engagement and motivation, as evidenced by both qualitative feedback and quantitative survey results. Fig. 4 illustrates the impact of the course on student motivation and its perceived effectiveness.

Fig. 4A shows a strong positive trend in the motivation and interest of students in chemical engineering and its industrial applications. 74% of the students reported high levels of increased motivation (scores of 4 and 5 on a 5-point scale), indicating the success of the course approach in fostering enthusiasm for the field.

The effectiveness of the PO-PBL methodology is further supported by the data in Fig. 4B, with 68.4% of the students finding it effective and an additional 10.5% rating it very effective. This strong endorsement underscores the success of the approach in captivating students' interest and creating a conducive learning environment. Fig. 4C reveals the preference of students for practical applications in the real world. 42.9% of the students valued solving real problems, while 38.1% appreciated the practical application of concepts. This emphasis on hands-on learning experiences aligns with PO-PBL's core principles and demonstrates its effectiveness in bridging theory and practice.

The practical nature of the course has enhanced the technical skills of the students and fostered critical thinking and problem-solving abilities. 82% of the students reported feeling challenged to adapt and find solutions in dynamic situations, reflecting the complexities of the real world. One student described this as “constantly adapting to problem-solving, like when something in the laboratory didn't work as expected” highlighting the course's success in cultivating resilience and adaptability.

The authentic, real-world challenges presented in the PO-PBL approach struck a chord with the students. 89% of students valued the opportunity to deal with problems that mirror those encountered in industry, offering a glimpse of their future professional lives. One student captured this sentiment, stating, “It has confronted me with a real problem, which is being able to sell an entrepreneurship idea.” This authenticity not only boosted engagement but also helped students frame their learning within broader career contexts.

In addition, the PO-PBL methodology has nurtured a culture of self-directed learning and research. 76% of students reported feeling driven to expand their knowledge beyond the course syllabus, developing fundamental skills for lifelong learning. As one respondent put it: “The use of application cases broadens my field of knowledge by making my research and acquiring knowledge outside the studied area.”

5.3. Communication skills development

The redesign of the POU course has markedly improved students'

communication skills, particularly in the written and visual domains. Fig. 5 illustrates the evolution of communication skills in four semesters from 2022 to 20–2024–10.

For written communication skills (Fig. 5A), the initial analysis revealed an upward trend in mean scores: from 3.335 in 2022–20–3.949 in 2023–10 and 3.932 in 2023–20, with a slight dip to 3.704 in 2024–10. The one-way ANOVA results (F statistic: 7.503, p-value: 0.00014) indicated statistically significant differences between semesters.

To address potential scoring bias, an impartial evaluation was conducted. Fig. 6 visualizes the results of this unbiased evaluation, comparing various aspects of written communication between the 2022–20 and 2024–10 semesters using six key criteria.

The visual representation corroborates the statistical findings, with t-test results for each criterion revealing significant improvements (p-values ranging from 0.005 to 0.014). These results demonstrate that the groups of the 2024–10 semester consistently outperformed their counterparts in the 2022–20 semester in all aspects of written communication.

Regarding visual communication skills (Fig. 5B), the one-way ANOVA (F-statistic: 3.843, p-value: 0.0105) indicates statistically significant differences between semesters. A general upward trend is observed, with mean scores increasing from 3.707 in 2022–20–3.944 in 2024–10. Tukey's HSD test reveals significant improvements between 2023 and 10 and both 2023–20 and 2024–10 semesters.

Qualitative feedback from students supports the effectiveness of these interventions. 87% of students reported significant improvements in presentation skills, with one student noting, “I feel much more confident in presenting technical information now. The course really helped me structure my thoughts and present them clearly to different audiences.” 79% of the students indicated a greater proficiency with graphical tools such as Power BI, and 83% reported an improved ability to create effective posters and flyers.

5.4. Professional skills development

The PO-PBL methodology implemented in the course has significantly influenced the development of various professional skills, with a particular focus on teamwork, creativity, innovation, and entrepreneurship. Fig. 7 illustrates the perceptions of students of the co-evaluation and self-evaluation processes throughout four semesters.

Fig. 7A depicts the distribution of the co-evaluation and self-evaluation scores from Semester 2022–20–2024–10. Consistently high median scores (above 4.5 on a 5-point scale) in all semesters suggest that students maintained a positive view of their teamwork skills throughout the course. Fig. 7B shows that 57.9% of students rated these evaluation methods as highly useful (score of 5), while 21.1% each gave ratings of 4 and 3.

Qualitative feedback further supports the course's emphasis on teamwork. One student remarked that the course “... has taught me many things associated with teamwork,” while another noted that it “increased my laboratory skills and allowed me to work and face challenges as a team.”

In terms of creativity and innovation, 84% of the students reported applying these competencies in various contexts, both within the course and in other academic pursuits. One student mentioned applying these skills “for the production of the patch and in subjects like reaction engineering,” while another utilized them “at the time of designing the patch when the emulsion was already available.”

Problem-solving and adaptation also stood out as key areas of skill application. 76% of students described employing “creativity and innovation in solving each time a problem arose and being able to move forward,” underscoring the course's role in fostering resilience and creative thinking when facing challenges.

5.5. Impact of GenAI tools

The integration of GenAI tools into the POU course has led to notable

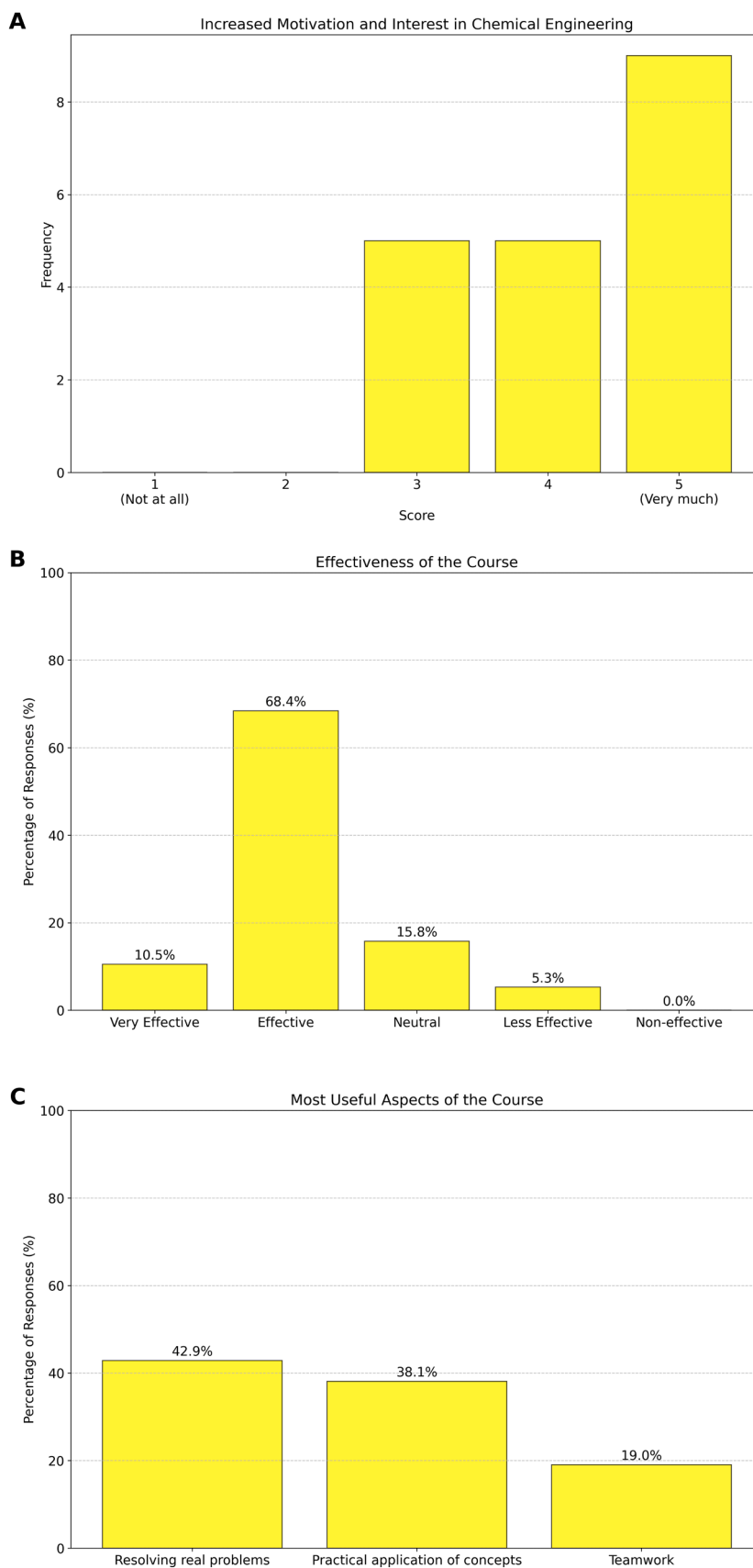


Fig. 4. Student survey results on PO-PBL methodology.

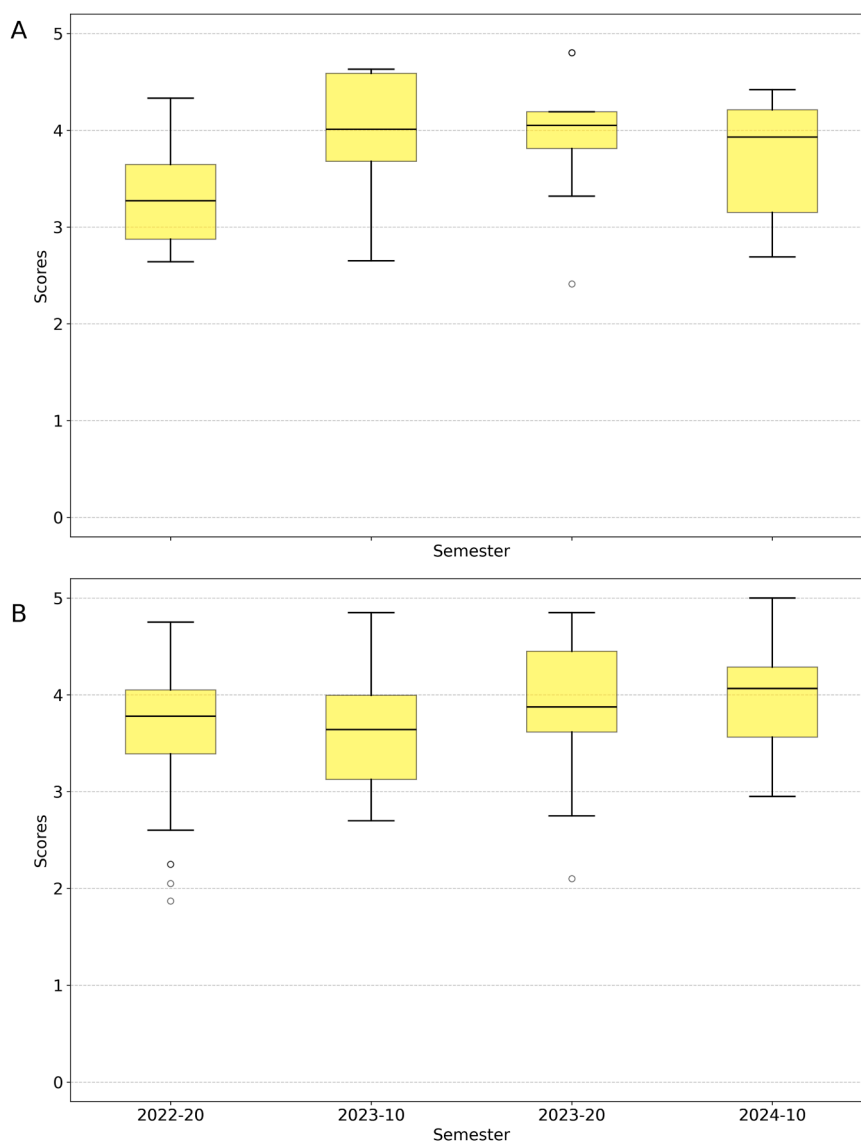


Fig. 5. Distribution of communication skills scores across semesters.

enhancements in students' research capabilities and writing skills. Fig. 8 illustrates substantial enhancements in several key aspects of literature utilization.

Fig. 8A reveals a marked increase in the average number of references used by students, rising from about 8 in the 2022–20 semester to more than 20 in the 2024–10 semester. This increase is statistically significant (t statistic: -2.579 , p-value: 0.045). The quality of the sources used also improved significantly, as shown in Fig. 8B. The number of sources with high impact nearly quadrupled, jumping from an average of about 3 in 2022–20–13 in 2024–10 (t statistic: -2.829 , p-value: 0.043).

The temporal distribution of references (Fig. 8C) shows a trend towards more recent publications in the 2024–10 semester, with a notable increase in citations from the past five years. Improvements in referencing quality are evident in Fig. 8D, which shows a substantial decrease in incomplete references from 2022 to 20–2024–10.

Students' perceptions of the utility of GenAI tools in literature review processes are overwhelmingly positive. Fig. 8E shows that 80% of the students rated the AI-assisted literature review instructions as highly useful (score of 5 on a 5-point scale), with the remaining 20% rating them as very useful (score of 4).

The influence of GenAI extends beyond literature review to ideation

and writing processes. Fig. 9 illustrates significant enhancements in methodology writing across three key criteria: *Design Clarity and Coherence*, *Properties Justification*, and *Properties Evaluation*.

Across the three areas, the 2024–10 cohort markedly outperformed the 2022–20 cohort. The average score for *Design Clarity and Coherence* rose from about 65–90%, *Properties Justification* from 65% to 88%, and *Properties Evaluation* from 65% to 89%.

5.6. Overall course effectiveness

The feedback system that professors and teaching assistants implemented has enhanced the learning process. 92% of students consistently praised the value of timely guidance, particularly during critical project phases. One student remarked, “they helped us structure the project and resolve doubts at critical moments,” highlighting the importance of targeted support.

The course has equipped students with a diverse set of skills applicable in various sectors of the chemical industry. 88% of the students reported acquiring practical skills needed for industry work, such as creating process flow diagrams (PFD), technical writing, and operating laboratory equipment.

Problem-solving and adaptability emerged as key competencies

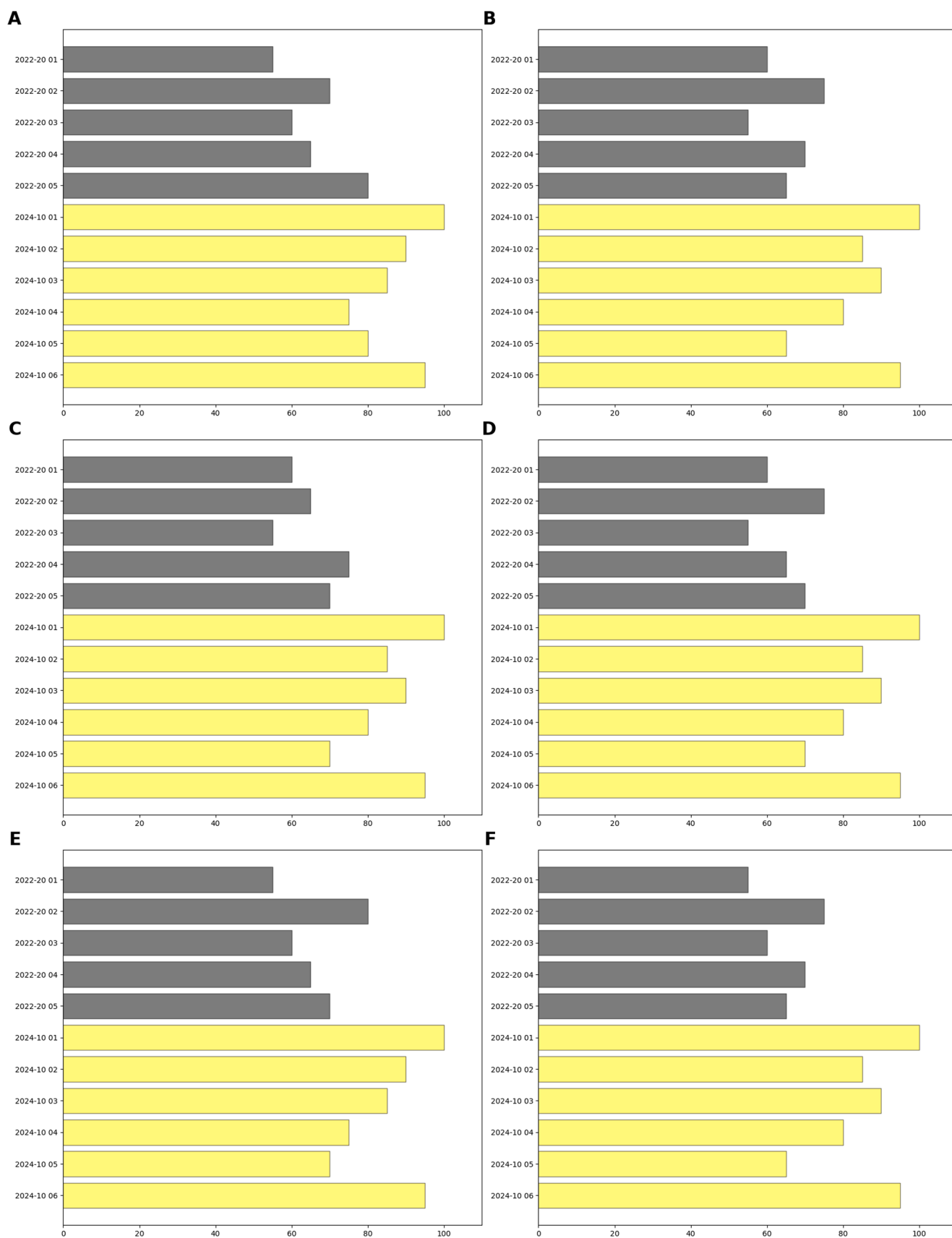


Fig. 6. Unbiased evaluation of group documents comparing 2022–20 and 2024–10 semesters.

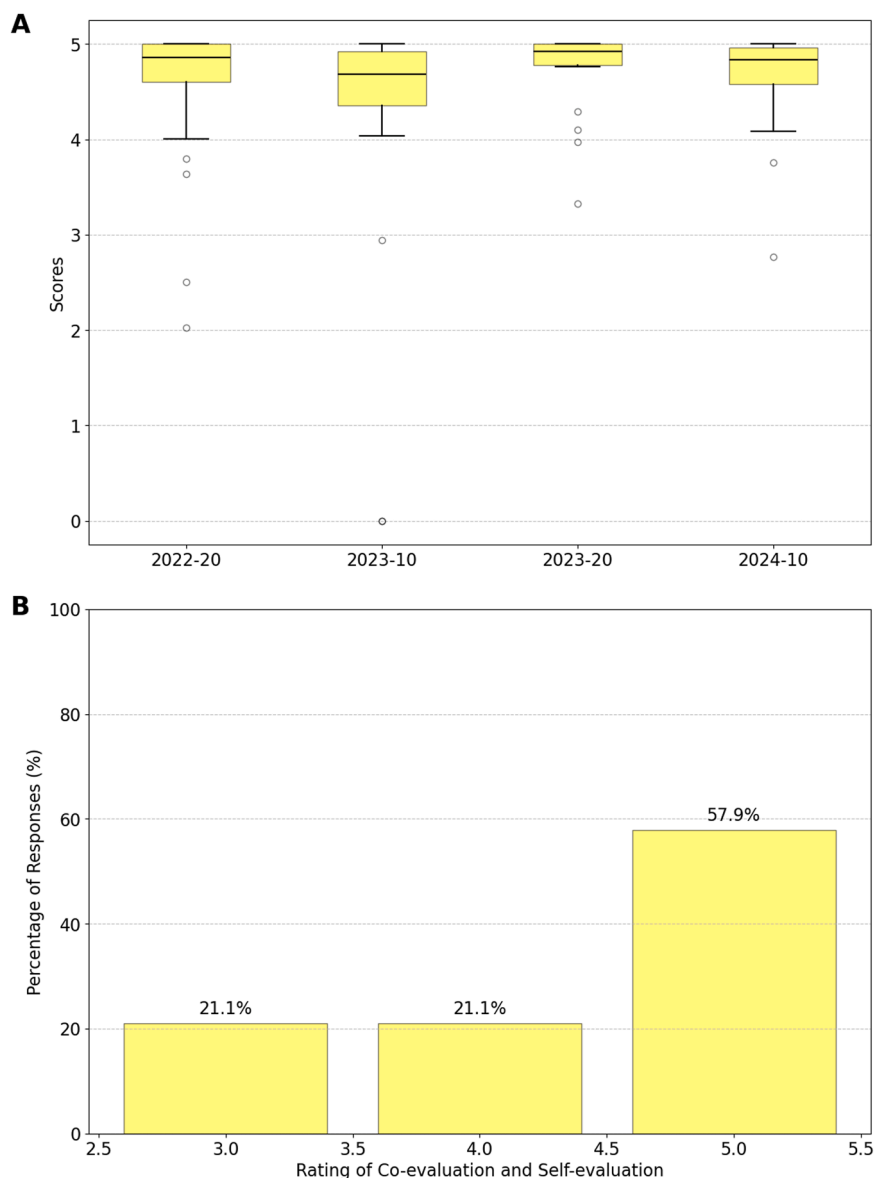


Fig. 7. Evaluation of teamwork skills.

honed throughout the course. 85% of the students felt better equipped to tackle new problems and develop viable solutions quickly. The course also imparted valuable industry-specific knowledge, particularly in areas like food and pharmaceuticals. 79% of the students appreciated learning about scaling processes from the laboratory to the industrial level.

The course's impact on personal and professional growth is evident in specific student testimonials. One student highlighted its influence on academic pursuits, stating, "thanks to the focus on innovation, I found the topic for my honors project." Another student emphasized the practical relevance, noting, "it has been very useful in providing me with tools for project development and experience in how projects would be developed in real life, in companies that might hire me."

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This feedback underscores the course's success in bridging academic

learning with industry-relevant skills. The development of resilience was highlighted by a student who mentioned overcoming specific challenges, such as "not being able to find the *Mitracarpus* (a plant the students tried to use to extract bioactive compounds)," illustrating how the course cultivates persistence in the face of real-world obstacles.

In general, 94% of the students reported feeling more prepared for their future careers in chemical engineering, with 87% indicating that the course had significantly influenced their career aspirations and understanding of the field.

6. Discussion

The implementation of Project-Oriented Problem-Based Learning (PO-PBL) in the Project of Unit Operations course demonstrates significant promise for advancing chemical engineering education. Our analysis in four semesters reveals several key findings that contribute to the broader discourse on engineering pedagogy.

The sustained improvement in technical skills, coupled with the enhanced engagement of the students, validates the effectiveness of our integrated approach. Students demonstrated consistent or improved understanding of unit operations while developing fundamental

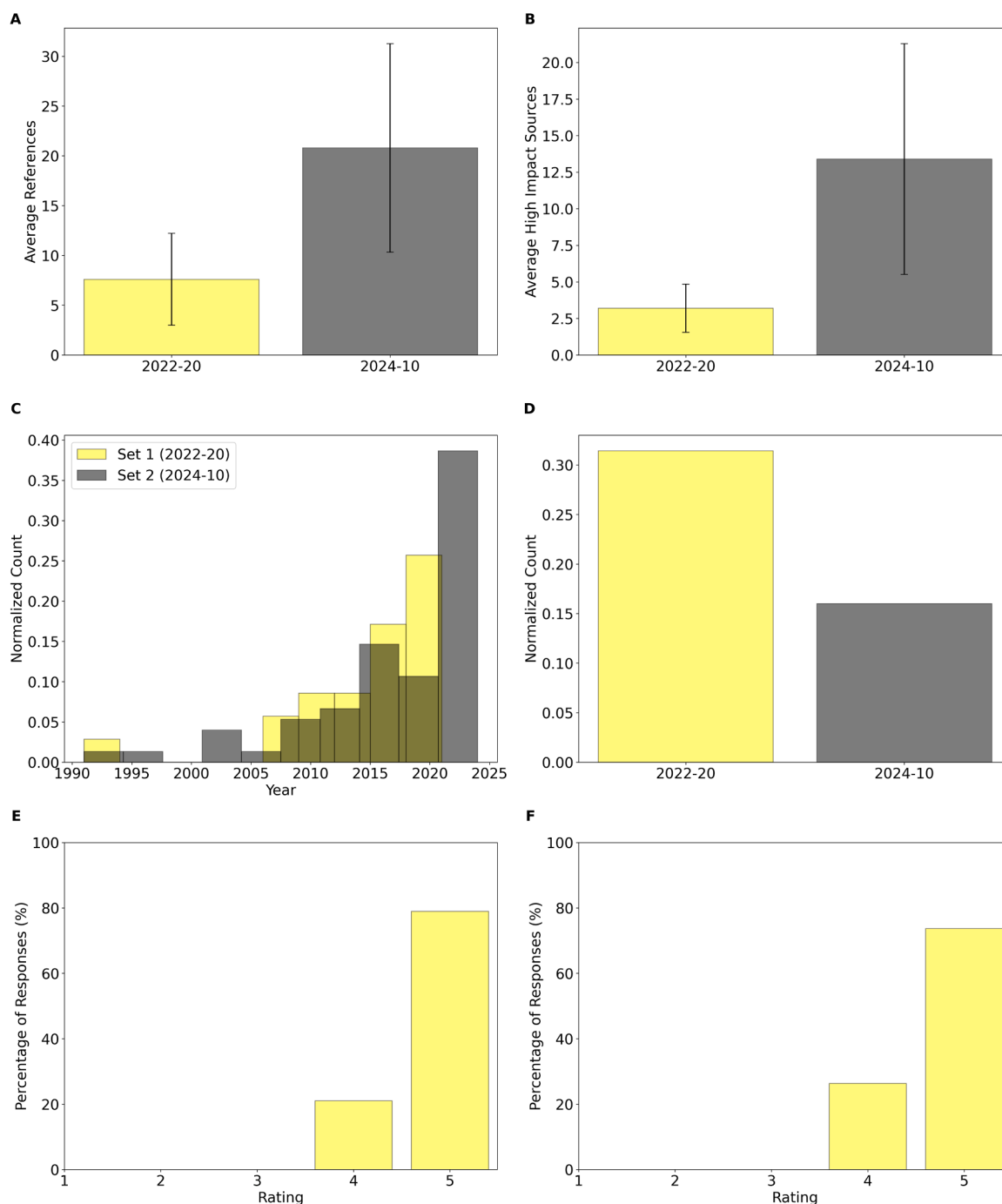


Fig. 8. Impact of GenAI tools on referencing and literature review.

professional competencies. The combination of PO-PBL with the DIPP methodology provided a structured framework that effectively bridged theoretical knowledge with practical application, addressing a long-standing challenge in engineering education.

The integration of GenAI tools, guided by the AIAS framework, significantly improved students' research capabilities and writing skills. Analysis of literature reviews showed a marked improvement in both quantity and quality of references, with the average number of citations increasing from 8 to over 20 and high-impact sources nearly quadrupling. This improvement suggests that properly guided AI integration can strengthen, rather than compromise, academic rigor.

Although the integration of GenAI tools has shown overall positive outcomes, our analysis reveals important nuances in its impact across

different students. The wider interquartile range in technical skills observed in the 2024–10 cohort (Fig. 3) suggests varying levels of effectiveness in the use of AI tools. This variability may indicate an emerging digital divide, where students' pre-existing technological proficiency influences their ability to leverage these tools effectively. Some students demonstrated exceptional ability to use GenAI tools to improve their learning, as evidenced by the performance of the upper quartile in technical skills assessments. However, others appeared to struggle with integrating these tools into their learning process, which could widen the achievement gap.

This observation highlights critical considerations for integrating AI-enhanced learning strategies into chemical engineering education. The effective use of GenAI tools hinges not only on their availability but also

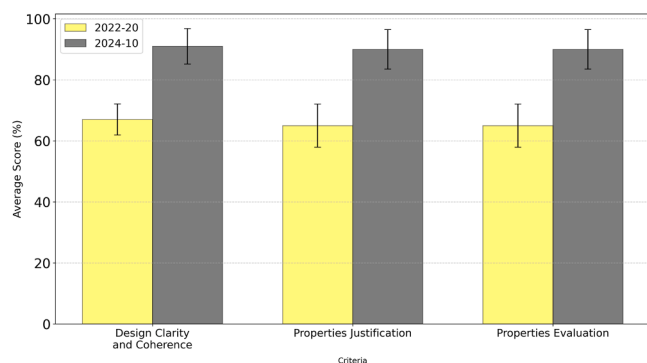


Fig. 9. Improvement in methodology writing across three criteria.

on students' proficiency in digital literacy and their capacity to critically evaluate AI-generated outputs. To address these challenges, the School of Engineering at the Universidad de los Andes has launched the Augmented Intelligence initiative, aimed at implementing GenAI strategies that complement rather than hinder the learning process. This institution-wide program includes the development of MOOCs focused on AI literacy, prompt engineering, ethical use, governance, and responsible integration of GenAI tools. By enhancing students' foundational competencies in these areas, we anticipate narrowing the observed skill gaps in subsequent course iterations. Future implementations of the POU course will benefit from this strengthened institutional support. Specifically, the redesigned curriculum will incorporate structured workshops, peer mentoring systems, and more granular monitoring of student engagement with AI tools. These measures are intended to ensure that all learners are equipped to effectively leverage emerging technologies, ultimately advancing their professional development within and beyond the classroom environment.

The development of communication skills emerged as a particular strength of our approach. The collaboration with the *Centro de Español* for written communication, combined with the focus of our teaching team on visual communication, resulted in measurable improvements in the ability of students to convey complex technical information. Statistical analysis confirmed a significant improvement in both written and visual communication skills in successive semesters.

The development of professional skills, particularly in teamwork and adaptability, showed consistent improvement. Student feedback and performance metrics indicate that the course's emphasis on real-world challenges and team-based problem-solving effectively prepared them for industry demands. The gamification elements successfully maintained engagement throughout the semester-long projects, with over 90% of students reporting increased motivation.

However, challenges remain in fully integrating theoretical concepts with practical applications. Student feedback indicates opportunities to strengthen the connection between laboratory sessions and project work. Future iterations of the course could benefit from more explicit links between theoretical workshops and practical challenges.

Looking ahead, our experience offers valuable insight for engineering educators looking to modernize their curricula. The success of this multifaceted approach, combining PO-PBL, DIPP methodology, GenAI tools, and professional skill development, suggests a promising direction for chemical engineering education. As we continue to refine this model, particular attention should be paid to the following.

- Further integration of theoretical concepts with practical applications
- Enhanced scaffolding of GenAI tool usage to maximize learning benefits
- Development of more sophisticated assessment methods for professional skills
- Expansion of industry collaboration opportunities

This study demonstrates that a well-structured PO-PBL approach, strengthened by contemporary tools and methodologies, can effectively equip chemical engineering students to navigate the complexities of professional practice without compromising academic rigor. The observed improvements in both technical and professional competencies suggest that this instructional model may serve as a valuable template for other institutions seeking to modernize their engineering curricula and better align them with evolving industry and societal needs.

7. Conclusion

The integration of Project-Oriented Problem-Based Learning (PO-PBL) in chemical engineering education, as exemplified by the Project of Unit Operations (POU) course, emerges as an effective approach to prepare students for the complexities of modern engineering practice. This method has shown particular strength in fostering fundamental professional skills alongside technical proficiency.

The sustained or enhanced technical skills observed throughout the semester, coupled with increased student engagement and motivation, underscore the effectiveness of the PO-PBL methodology. The course has made significant strides in addressing critical aspects of engineering education, particularly in communication skills and professional skill development.

The integration of GenAI tools, guided by the Artificial Intelligence Assessment Scale (AIAS), has significantly enhanced students' research capabilities and writing skills. This integration prepares students for a future in which artificial intelligence will play an increasingly significant role in engineering practice.

As we continue to refine and adapt our educational methodologies, the principles and practices demonstrated in the POU course offer valuable insights to educators across engineering disciplines. By embracing a more holistic and practical approach to education, we can create learning environments that not only impart technical knowledge but also nurture the creativity, resilience, and adaptability essential for the next generation of chemical engineers to flourish in their careers and contribute meaningfully to society.

CRedit authorship contribution statement

Angie Daniela Bolanos-Barbosa: Methodology, Investigation; **Julian Daniel Torres-Vanegas:** Methodology, Investigation; **Deya Marcela Santos Carvajal:** Methodology, Investigation; **Luis H. Reyes:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Oscar A Alvarez Solano:** Writing – review & editing, Validation, Conceptualization; **Juan C Cruz:** Writing – review & editing, Writing – original draft, Visualization, Formal analysis, Conceptualization; **Olga Lucía Acuña:** Methodology, Investigation

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

The authors utilized the generative AI tool, the Anthropic Claude 3.5 Sonnet, for grammar correction and text editing during the preparation of this manuscript. Following the use of this tool, the authors thoroughly reviewed and edited the content to ensure its accuracy and quality. The authors take full responsibility for the final content of the publication.

Conflict of Interest

The authors declare that there are no conflicts of interest to disclose related to the research, authorship, or publication of this manuscript.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.ece.2025.01.001.

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Glossary

- 3P Tool:** A multiscale approach used in the DIPP methodology to link product formulation with macroscopic, microscopic, and molecular properties and process conditions.
- AIAS:** (Artificial Intelligence Assessment Scale) A framework providing guidelines for the ethical and effective use of AI in educational contexts.
- Antibiogram:** A test used to determine the sensitivity or resistance of bacteria to specific antibiotics.
- Centro de Español:** (Spanish Center) A unit at Universidad de los Andes dedicated to enhancing students' communication skills in academic and professional contexts through interdisciplinary strategies and support services.
- ChatGPT:** An AI-powered language model developed by OpenAI, capable of generating human-like text based on input prompts.
- DIPP:** (Diseño Integrado de Producto y Proceso) Spanish for Integrated Product and Process Design, a methodology that merges product design with process engineering.
- FTIR:** (Fourier Transform Infrared Spectroscopy) An analytical technique used to identify organic, polymeric, and, in some cases, inorganic materials.
- GenAI:** Generative Artificial Intelligence refers to AI systems that create new content, such as text, images, or data.
- Generation Z:** (Gen Z) The demographic cohort succeeding Millennials, typically defined as those born from the mid-to-late 1990s to the early 2010s.
- PBL:** (Problem-Based Learning) An educational approach in which students learn about a subject through the experience of solving open-ended problems.
- PFD:** (Process Flow Diagram) A diagram commonly used in chemical and process engineering to indicate the general flow of plant processes and equipment.
- PO-PBL:** (Project-Oriented Problem-Based Learning) An educational approach that combines project-based learning with problem-based learning, focusing on long-term, complex problems or challenges.
- POU:** (Project of Unit Operations) The specific course discussed in this article, which implements PO-PBL in chemical engineering education.
- Power BI:** A Microsoft business analytics tool for interactive visualizations and business intelligence capabilities.
- Psychological safety:** In a team setting, the shared belief that the team is safe for interpersonal risk-taking allows members to speak out without fear of negative consequences.
- Pugh matrix:** A decision-making tool used to evaluate and prioritize potential solutions to a problem based on multiple criteria.
- SWOT analysis:** A strategic planning technique used to help identify Strengths, Weaknesses, Opportunities, and Threats related to business competition or project planning.
- Unit Operations:** The basic physical operations of chemical engineering in a chemical process plant, such as filtration, distillation, or heat exchange.